

HARAMRIT SINGH KHURANA

Grapevine, TX · (703) 341-9667 · haramrit09k@gmail.com · github.com/haramrit09k · linkedin.com/in/haramrit09k

SUMMARY

Senior Software Engineer (AVP) at Citi since Dec 2022, building and operating backend systems for an internal MongoDB DBaaS platform (~2,500 clusters, 600 daily users, 7,000–8,000 API requests/day) used by application teams across the firm: Java/Spring Boot microservices, REST/GraphQL APIs, and asynchronous workflow orchestration. Flagship GenAI work: an LLM-assisted diagnostics agent on an internal enterprise framework based on Google ADK with MCP tooling, read-only constraints, and human-in-the-loop design.

EXPERIENCE

Citibank Dec 2022 – Present
Senior Software Engineer (AVP) *Texas*

- Built and operated features for Citi's internal MongoDB DBaaS platform (~2,500 clusters, 600 daily users, 7,000–8,000 API requests/day); owned production releases across microservices serving many internal application teams.
- Built an LLM-assisted MongoDB diagnostics workflow on an internal enterprise framework based on Google ADK: the LLM plans diagnostic actions and invokes tools through a Mongo-focused MCP server against Apigee-backed APIs to produce targeted database checks and grounded remediation guidance — read-only constraints, human-in-the-loop design.
- Redesigned a high-volume metadata ingestion workflow (~49K records × 150+ fields, ~300 MB payload), reducing peak JVM heap usage from ~1.4 GB to ~470 MB via streaming and efficient data handling.
- Optimized a production-critical Oracle-backed workflow, cutting execution-plan cost from ~20,000 to ~370 and response times under 10 seconds; deep SQL/query-diagnosis background (Oracle, SQL Server, MySQL).
- Automated ServiceNow CHG creation for 3 critical MongoDB operations, deriving required change metadata from platform context to eliminate repetitive manual ticket preparation while preserving approval controls.
- Automated CyberArk FID migrations after cluster topology/capacity changes: asynchronous job tracking, hourly polling with on-demand refresh, per-account status visibility, and escalation only for failures.

Optimal Satcom 2021 – 2022
Software Engineer

- Modernized legacy .NET Core/Angular capacity-management applications and reengineered failure-prone backend modules; optimized complex SQL stored procedures and triggers.

PROJECTS

- AI Interview Prep Mentor** | *React, TypeScript, Express, Gemini, PostgreSQL, Redis*
Full-stack app using Gemini to generate interview questions and AI-driven evaluation/feedback; PostgreSQL in production, Redis caching, auth, and payments.
- F1rstAid** | *Python, LangChain, OpenAI, FAISS, Streamlit*
RAG assistant over government and university immigration sources, with knowledge-base refresh and lint/test infrastructure.
- LogScribe MCP** | *Python, MCP, Claude Desktop, Pytest*
MCP server exposing log-analysis tools (search, filtering, analytics); integrated with Claude Desktop; tested with Pytest.

TECHNICAL SKILLS

Languages: Java, SQL, TypeScript; C#/.NET Core, Python (tooling and GenAI project work), JavaScript

Backend: Spring Boot, REST/GraphQL API design, microservices, JDBC, asynchronous and long-running workflow orchestration, enterprise API integrations

Databases: MongoDB DBaaS platform (~2,500 clusters), MongoDB Ops Manager APIs; Oracle, SQL Server, MySQL — query optimization and production debugging; PostgreSQL in production project work

Platform: OpenShift, Docker, Kubernetes concepts, Jenkins CI/CD, Playwright, Angular, production release ownership

AI: LLM application development, RAG, MCP tooling, human-in-the-loop design, internal enterprise framework based on Google ADK, Python tooling (LangChain, FAISS)

EDUCATION & PUBLICATION

M.S., IT Project Management — New England College 2026

M.S., Computer Science (GPA 3.75) — George Mason University 2021

B.E., Computer Engineering — University of Mumbai 2019

- A Resource Effective Approach for Distributing Machine Learning Models over a Cluster** — ICAC3 2019 (IEEE)